ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** hypotonic to the cell	Water entered.
3	**(a)** synthesis of ATP	Mitochondria do ATP.
4	**(a)** cell wall	Rigid wall.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** hypertonic to A and hypotonic to B	Different internal concentrations.
7	**(c)** bacterial	Prokaryote.
8	**(b)** large central vacuole	Takes up space.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** 10	Mitosis keeps 20; meiosis halves.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1).	1+1
12	It protects the cell and controls entry and exit of substances (1); It allows only some substances to pass (1).	1+1
13	Protoplasm separates from the cell wall (1); Gap between wall and membrane (1).	1+1
14	4 cells (1); 46 chromosomes in each (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1).	1+1+1
16	Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1).	1+1+1
17	Fill the bag with starch and glucose (1); Place in water (1); Glucose passes out; starch stays (1).	1+1+1
	OR: Root tip: mitosis, growth (1); Testis: meiosis for gametes (1); Chromosome numbers (1).	1+1+1
18	They contain chlorophyll (1); They trap sunlight (1); They make food (glucose) by photosynthesis (1).	1+1+1
	OR: Ability of a cell to form a whole organism (1); Plant cells are totipotent (1); Used to grow plants from single cells (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cut equal potato cubes, weigh (1); Place in solutions for same time (1); Reweigh and calculate the percentage change (1); Plot the graph of percentage change vs concentration (1); Intercept on the x-axis gives isotonic concentration (1).	1+1+1+1+1
	OR: Stem cells unspecialised (1); Differentiation into RBC, nerve, muscle (1); Plant meristems (1); Uses: bone marrow transplants (1); Ethical issues: source of embryonic cells (1).	1+1+1+1+1
20	Prokaryotic: no true nucleus, no membrane-bound organelles (1); Eukaryotic: nucleus and organelles (1); Eukaryotic cells divide the work among organelles (1); Viruses have no cell structure (1); Viruses multiply only inside host cells (1).	1+1+1+1+1
	OR: Mammalian red blood cell (1); Biconcave disc, large surface (1); More space for haemoglobin (1); No nucleus means no repair or division (1); Short life of about 120 days (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** energy release	Respiration.
2	**(c)** vacuole	Water and solutes.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 16	2^4.
5	**(a)** proteins	Protein synthesis.
6	**(d)** petals and ripe fruits	Pigments.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** 8	2 x 2 x 2.
9	**(c)** root cells lack chloroplasts	Light and chloroplasts.
10	**(a)** potato tubers	Store starch.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mitochondria (1); Cellular respiration (1).	1+1
12	Plant cells have a cell wall (1); Animal cells do not (1).	1+1
13	Makes ribosomal subunits (1); Ribosomes make proteins (1).	1+1
14	Enzymes digest the cell (1); Autolysis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mitosis: 8 in each cell (1); Meiosis I: 4 chromosomes (each with two strands) in two cells (1); Meiosis II: 4 chromosomes in four cells (1).	1+1+1
16	Tiny channels in the cell wall (1); Connect cytoplasm of adjacent cells (1); Exchange of substances (1).	1+1+1
17	For growth (1); For repair of damaged tissues (1); For replacement of old cells (1).	1+1+1
	OR: Mature cells store water and maintain turgor (1); Meristem cells are dividing and dense (1); Vacuoles enlarge on maturity (1).	1+1+1
18	Volume ratio = (25)^3 = 15625 (1); SA/V is inversely proportional to the diameter (1); Bacterium has 25 times larger SA/V (1).	1+1+1
	OR: Proteins are not packaged (1); Not sent out by vesicles (1); Secretion fails (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	DNA in nucleus transcribed (1); Ribosomes on rough ER translate (1); Golgi modifies and packs (1); Vesicle fuses with the membrane (1); Mitochondria supply ATP (1).	1+1+1+1+1
	OR: SA/V = 6, 3, 1.5 (1); Ratio falls with size (1); Large organisms need lungs, gills (1); Circulatory systems carry materials (1); Thin and folded surfaces increase exchange (1).	1+1+1+1+1
20	Water enters by osmosis (1); Turgor pressure builds up (1); Wall pressure balances (1); Equilibrium: no net movement (1); Protoplast without wall bursts (1).	1+1+1+1+1
	OR: Equal discs washed (1); Water at different temperatures (1); Measure colour intensity or absorbance (1); Greater leakage at higher temperatures (1); Heat damages proteins and lipids of the membrane (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 2 : 1	46 : 23.
3	**(a)** turgor in cells	Osmosis.
4	**(b)** leucoplast	Amyloplast.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** half	SA/V = 6/a.
7	**(a)** heat denatures the membrane proteins	Leaky membrane.
8	**(a)** no membrane-bound organelles	Simple cell.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** 8 times	Cube.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rapid production of identical plants (1); Disease-free plants (1).	1+1
12	Underground, no light (1); They are growing cells (1).	1+1
13	Osmosis is passive, active transport needs energy (1); Osmosis is along gradient; active is against it (1).	1+1
14	Division of work (1); Increases efficiency by compartments (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus gives mRNA (1); Ribosomes on rough ER make protein (1); Golgi packs; vesicles go to the membrane (1).	1+1+1
16	Initially isotonic (1); Solute may enter or water may evaporate (1); Concentration changes (1).	1+1+1
17	High solute outside (1); Microbes lose water (1); Cannot multiply (1).	1+1+1
	OR: No cell structure (1); Cannot reproduce on their own (1); Bacteria have ribosomes, membranes, own metabolism (1).	1+1+1
18	Epidermis protects (1); Palisade packed with chloroplasts for light (1); Structure related to function (1).	1+1+1
	OR: Small organism: short distances, high SA/V (1); Large body: long distances (1); Needs circulatory and respiratory systems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Number of divisions: one vs two (1); Number of daughter cells: two vs four (1); Chromosome number: same vs halved (1); Occurs in: body cells vs reproductive organs (1); Significance: growth and repair vs formation of gametes (1).	1+1+1+1+1
	OR: Hooke: cells in cork (1); Leeuwenhoek: microorganisms (1); Brown: nucleus (1); Schleiden, Schwann, Virchow: cell theory (1); Electron microscope: organelles (1).	1+1+1+1+1
20	Common: nucleus, mitochondria, ER, Golgi, ribosomes, lysosomes (1); Plant only: wall, plastids, large vacuole (1); Animal only: centrioles (1); Structure of each (1); Function of each (1).	1+1+1+1+1
	OR: Protects the cell contents (1); Selective permeability controls entry and exit (1); Receptors on the surface receive signals (1); Keeps concentrations inside stable (1); Hence cell survives in changing surroundings (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.